MA378 Chapter 3: Numerical Integration

§3.1 Introduction / Newton-Cotes / The Trapezium Rule

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1.1 Introduction

Problem

Given a real-valued function f that is continuous on $[a, b]$, can we find an estimate for

$$I(f) := \int_a^b f(x)dx?$$

And if we can, can we say how accurate it is?

$I(\cdot)$ is the definite integral operator.

$\int_a^b f(x)\,dx$ is the area between $x=a$, $x=b$, $y=0$ and $y=f(x)$. "Area under the curve".

It is a real number.

1.1 Introduction

Why is this an interesting problem?

- Many problems in applicable mathematics require definite integrals to be evaluated. These methods were originally motivated by problems in astronomy, and brewing. They are now ubiquitous.
- Evaluating them by finding the anti-derivative can be hard, and very hard to automate.
- Some times, although the function is integrable, its anti-derivative doesn't exist in a closed form.

1.1 Introduction

The process of numerically estimating a definite integral is called **Numerical Integration** or **Quadrature**.

The formulae we'll derive all look like

$$Q_N(f) := q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + \cdots + q_N f(x_N).$$

Here the points x_i are called **quadrature points** and the q_i are **quadrature weights**.

We need a way of choosing these.

The simplest approach for choosing the quadrature points to take them to be equally spaced, i.e., $x_i = a + hi$ where $h = (b - a)/N$.

1.1 Introduction

How to choose the weights?

We've spent a lot of time on approximating functions with polynomials. So it is natural to compute a polynomial interpolant to f, and take its integral. The appeal of this approach is due to the fact that

- We know how to compute polynomial interpolants.
- Integrating polynomials is easy.
- We can estimate the error easily (yet again, we'll make use of Cauchy's Theorem).

This leads to the **Newton-Cotes** methods, which are the subject of this section, and the next one. Later again, we'll look at more sophisticated methods, called **Gaussian Methods** which use non-uniformly spaced points.

1.2 Newton-Cotes methods

Definition 1.1 (Newton-Cotes quadrature)

The **Newton-Cotes** quadrature rule for $\int_a^b f(x)dx$ with $N+1$ points is derived by integrating exactly the polynomial of degree $\cancel{n}$ N that interpolates f at the $N+1$ equally spaced points $a = x_0 < x_1 < \cdots < x_N = b$. The method is written as

$$Q_N(f) := q_0 f_0 + q_1 f_1 + q_2 f_2 + \cdots + q_N f_N,$$

where we use the notation $f_k := f(x_k)$.

That is, the quadrature weights are chosen so that

$$Q_N(f) = \int_a^b p_N(x)dx,$$

where p_N is the polynomial of degree n that interpolates f at the $N+1$ quadrature points...

1.2 Newton-Cotes methods

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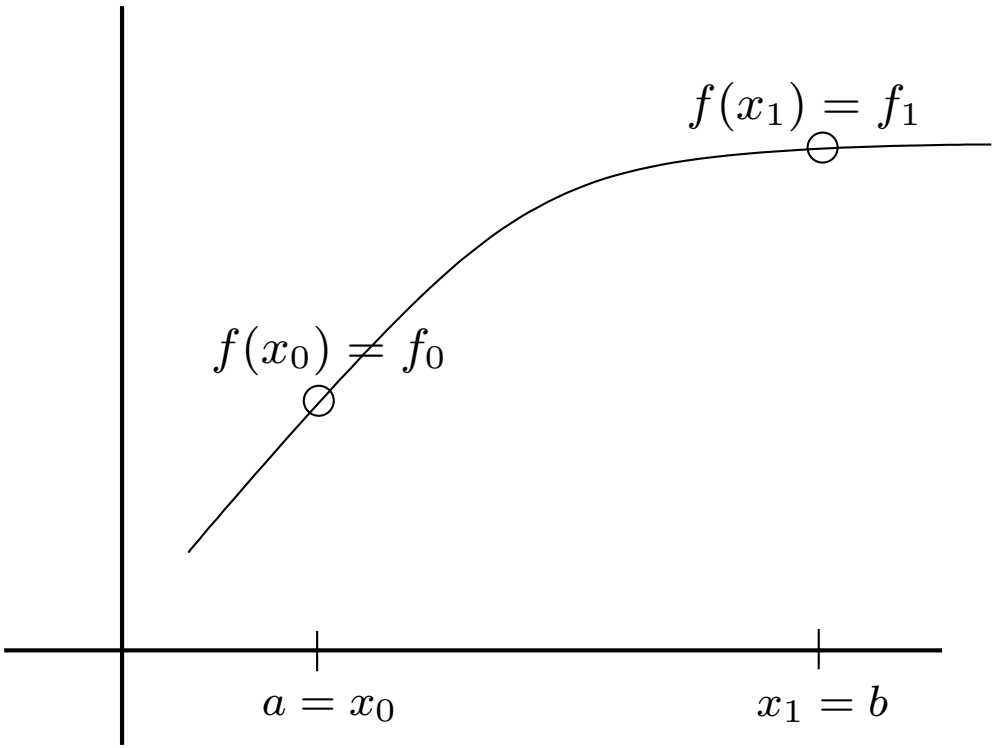
However, it turns out that we can compute the weights q_0, q_1, ..., q_N, **without** knowing p_N.

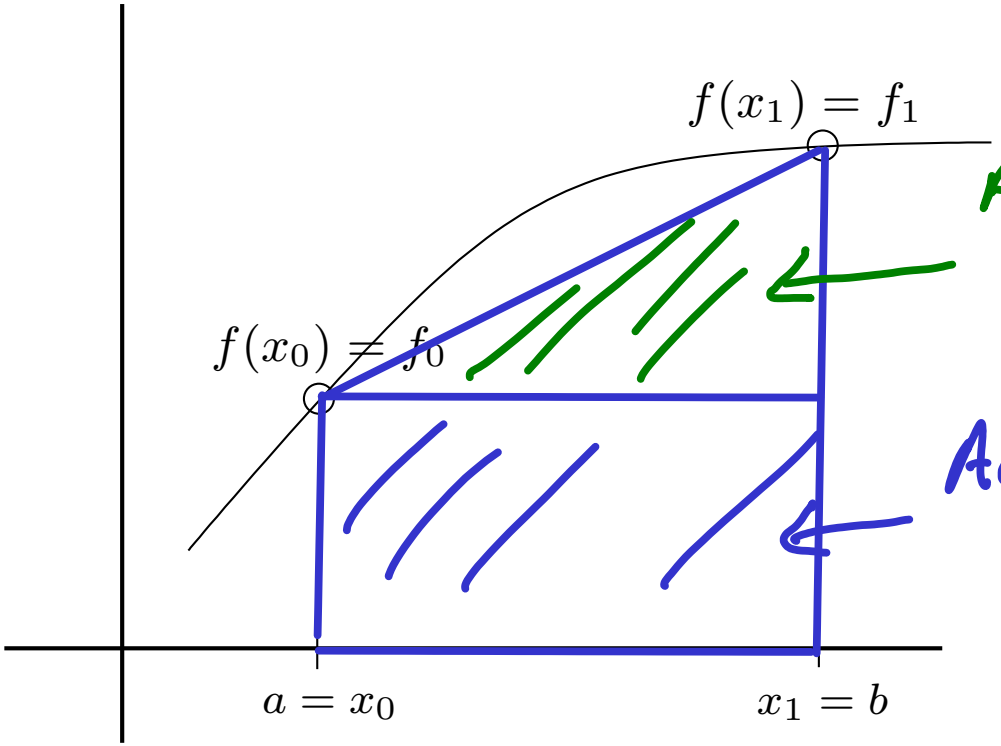
We'll do this for $N = 1$ in the next section, and $N = 2$ (the most interesting case) in Section 3.2.

1.3 The Trapezium rule

Suppose we wanted to estimate the integral of a function, f, shown below, on the interval $[a, b]$.



Method 1: We could try to estimate the area of the trapezium that fits under the graph:



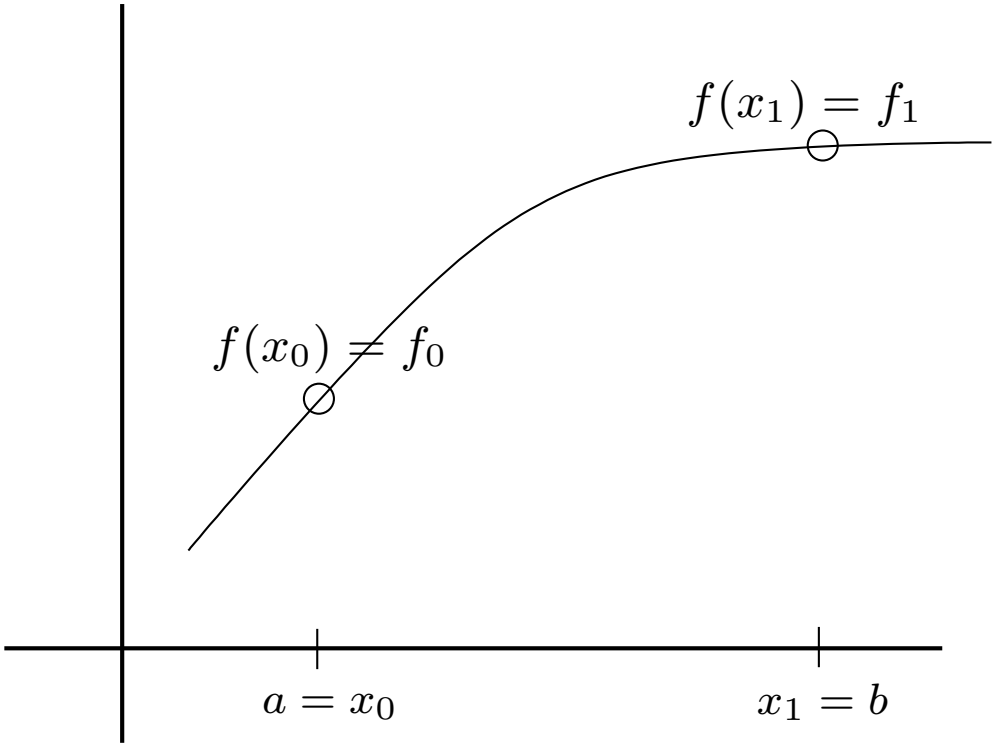
Area is $\frac{1}{2}(x_1 - x_0)(f_1 - f_0)$

Area is $(x_1 - x_0) f_0$

Area of Trapezium is

$$(x_1 - x_0) f_0 + (x_1 - x_0)\left(\frac{f_1}{2} - \frac{f_0}{2}\right) = \frac{1}{2}(f_0 + f_1)(x_1 - x_0)$$

Method 2: We could find p_1, the polynomial of degree 1 that interpolates f at $x = a$ and $x = b$:



Note that this shows that $q_i = \int_a^b L_i(x)dx$, where, as usual, the L_i are the Lagrange Polynomials.

Method 3: The third approach for generating the Trapezium Rule is called the *Method of Undetermined Coefficients*. Because the method is based on integrating a linear function we expect it to yield an exact solution for any constant or linear function (i.e., there should be no error). To keep the algebra simple, we'll take $a = 0$ and $b = 1$. So,

$$Q_1(f) = q_0 f(0) + q_1 f(1),$$

and, setting $f(x) \equiv 1$, and then $f(x) = x$ we get

1.3 The Trapezium rule

Method 3

Now we need to extend this to estimating $\int_a^b g(x)dx$ as follows:

Example 1.2

Use the trapezoid to estimate

$$\int_0^{\pi/4} \cos(x)dx.$$

Calculate the (exact) error $|\int_a^b f(x)dx - Q_1(f)|$.

1.4 Exercises

Exercise 1.1

(*For simplicity, you may assume that the quadrature rule is integrating f on the interval* $[-1, 1]$.) Let q_0, q_1, ..., q_n be the quadrature weights for the Newton-Cotes rule $Q_n(f)$. Show that $q_i = q_{n-i}$ for $i = 0, \ldots n$.

Exercise 1.2

Show that $\sum_{i=0}^{n} q_i = b - a$.